

# Doudou Cao 曹豆豆

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[Institute profile](#) | [Personal website](#) | [LinkedIn](#) | [Google Scholar](#)

## RESEARCH PROFILE

I am a biological anthropologist and osteoarchaeologist studying how ancient populations experienced health, growth, violence, and adaptation across environmental and historical change. My research combines human skeletal and morphometric data with environmental, spatial, and archaeological evidence to model long-term biological variation and population processes in ancient China. I use osteological analysis, 3D digital methods, spatial analysis, Bayesian modelling, and machine-learning approaches to examine interactions among human biology, environmental pressure, and social organisation, with case studies ranging from the Tibetan Plateau and southwest China to broader comparative datasets across China.

## RESEARCH INTERESTS

- Biological anthropology, osteoarchaeology, and human skeletal biology
- High-altitude adaptation, growth and development, and health in past populations
- Human body size and proportions across ecological and historical contexts
- Demography, inequality, violence, and social change in archaeological populations
- Quantitative and computational archaeology; Bayesian inference and spatiotemporal modelling
- Commingled human remains, secondary burial, and mortuary formation processes

## ACADEMIC APPOINTMENTS

2026-present      **Research Assistant Professor, Institute for the Humanities and Social Sciences, The University of Hong Kong**  
Research in biological anthropology, osteoarchaeology, quantitative/computational archaeology, high-altitude adaptation, and demography and health in past populations.

## EDUCATION

2021-2025  
Cambridge, UK      **PhD in Archaeology / Biological Anthropology, University of Cambridge**  
Department of Archaeology. Dissertation research on high-altitude adaptation, human body size and proportions, and comparative skeletal variation in ancient Tibetan and lowland Chinese populations.

2018-2021  
Chengdu, China      **MA in Archaeology, Sichuan University**  
School of Archaeology and Museology. GPA: 3.8/4.0. Thesis: Research on human skeletal remains from the Phiyang-Dungkar tombs in Western Tibet.

2014-2018  
Chengdu, China      **BA in Archaeology, Sichuan University**  
School of Archaeology and Museology. GPA: 3.6/4.0.

## MAJOR AWARDS AND HONOURS

2021-2025      **Harding Distinguished Postgraduate Scholars Programme, University of Cambridge**  
Full PhD scholarship and research support.

2022      **International Committee of the Red Cross (ICRC) Research Scholarship**  
Research scholarship supporting work related to missing persons and unidentified human remains.

2018-2021      **National Scholarship for Master's Degree, China**  
First Class.

2020      **Outstanding Student Leader Award, Graduate School, Sichuan University**

2018      **Outstanding Undergraduate Dissertation, Sichuan University**

2018      **China Guardian & Xu Bangda Art Scholarship**

2017      **Outstanding Undergraduate Student Award, Sichuan University**

## PUBLICATIONS

### Journal articles

- Cao, D., Crema, E. R., & Pomeroy, E. (2026). Multifactorial drivers of spatial variation in human body form across modern China. *Scientific Reports*. <https://doi.org/10.1038/s41598-026-57537-8>
- Wang, X., Yang, A., Yang, F., Yuan, H., Cao, D., Li, J., Huo, W., Lu, S., & Li, S. (2026). Subsistence change in Iron Age to Tubo-period western Xizang: A stable isotope study from Phiyang Dunkar. *Journal of Archaeological Science: Reports*, 70, 105639. <https://doi.org/10.1016/j.jasrep.2026.105639>
- Cao, D., Liu, X., Gu, W., Zhu, H., Zhang, R., Zhou, Z., Wei, Q., Zou, J., Qiu, Y., Chen, J., Ding, L., Pomeroy, E., & Yuan, H. (2025). Dental caries as indicators of agricultural practices in the foothills of Neolithic China. *Archaeological Research in Asia*, 43, 100645. <https://doi.org/10.1016/j.ara.2025.100645>
- Wang, X., Cao, D., Gu, W., Wei, Q., Chen, X.-L., Zhang, R., Ding, L., Tang, Z., Teng, X., Wang, T., Zou, J., Qiu, Y., Fu, Q., & Yuan, H. (2025). Multi-isotope analysis reveals human millet-based diets and limited mobility in the Central Plains of China ca. 5000 years ago. *International Journal of Osteoarchaeology*, 35, e3407. <https://doi.org/10.1002/oa.3407>
- Cao, D., Crema, E. R., & Pomeroy, E. (2024). Estimating intralimb proportions for commingled remains. *International Journal of Osteoarchaeology*. <https://doi.org/10.1002/oa.3326>
- Yi, B., Yuan, H., Wang, T., Cao, D., Yang, Y., & Hu, Y. (2024). Life history of a high-status noblewoman from the late Shu state during the Eastern Zhou period (770-221 BC): Childhood stress and adult stability. *International Journal of Osteoarchaeology*, 34(1).
- Tang, L., Wilkin, S., Richter, K. K., et al. (2023). Paleoproteomic evidence reveals dairying supported prehistoric occupation of the highland Tibetan Plateau. *Science Advances*, 9, eadf0345. <https://doi.org/10.1126/sciadv.adf0345>
- Tao, L., Yuan, H., Zhu, K., Liu, X., Guo, J., Min, R., He, H., Cao, D., Yang, X., Zhou, Z., Wang, R., Zhao, D., Ma, H., Chen, J., Zhao, J., Li, Y., He, Y., Suo, D., Zhang, R., Li, S., Li, L., Yang, F., Li, H., Zhang, L., Jin, L., & Wang, C.-C. (2023). Ancient genomes reveal millet farming-related demic diffusion from the Yellow River into southwest China. *Current Biology*, 33(22), 4995-5002.e7. <https://doi.org/10.1016/j.cub.2023.09.055>
- Lu, H., Li, Z., Ciren, C., Cao, D., Gao, X., & Li, L. (2022). Sding Chung: An early multi-burial cave site on the Tibetan Plateau. *Antiquity*, 96(387), 745-753. <https://doi.org/10.15184/aqy.2022.42>
- Yuan, H., Gu, W., Wei, Q., Wu, Q., Ding, L., & Cao, D. (2022). Analysis of dental caries in the Yangshao population at the Qingtai site, Zhengzhou. *Acta Anthropologica Sinica*, 41(2), 226-237. [in Chinese] <https://www.anthropol.ac.cn/EN/10.16359/j.1000-3193/AAS.2021.0023>
- Cao, D., & Yuan, H. (2021). Preliminary analysis of commingled human remains. *Yellow River, Yellow Earth, Yellow Race*, (06), 41-44. [in Chinese]

### Newsletter and thesis

- Cao, D. (2017, November 17). Seeing humans through bones: A multidimensional perspective on ancient human life - Summary of the Second Annual Meeting of the Osteoarchaeology Committee of the Chinese Archaeological Society. *China Cultural Relics News*, (006). [in Chinese]
- Cao, D. (2021). Research on human skeletal remains from the Phiyang-Dungkar tombs in Western Tibet. Master's thesis, Sichuan University, Chengdu, China. [in Chinese] <https://kns.cnki.net/KCMS/detail/detail.aspx?dbname=CMFDTEMP&filename=1022445375.nh>

## CONFERENCE PRESENTATIONS

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| 2026 | <b>Cao, D., Lü, H., Yang, F., Li, S., Yuan, H., Ciren, C., Wangdue, S., He, Y., Li, Y., Liu, X., Ji, K., Zou, J., &amp; Pomeroy, E. Adaptation or constraint? Skeletal evidence for growth and health in Himalayan highland populations. Trans-Eurasian Highlands: Insights from Multidisciplinary Archaeological Research, Center for Archaeological Science, Sichuan University, Chengdu, China, 9-13 March 2026. (Podium)</b> |
| 2026 | <b>Cao, D., Pomeroy, E., Lü, H., Yuan, H., Yang, F., &amp; Li, S. Bayesian modelling reveals spatiotemporal patterns of body size variation in Holocene Chinese populations. The Second Early China Development and Its Long-Term Impact Workshop, The University of Hong Kong, Hong Kong. (Podium)</b>                                                                                                                          |
| 2025 | <b>Cao, D., Crema, E. R., Lü, H., Yuan, H., &amp; Pomeroy, E. Tracking spatial and temporal variation in body form across ancient Chinese populations. Annual Conference of Computer Applications and Quantitative Methods in Archaeology (CAA UK), Cambridge, United Kingdom. (Podium)</b>                                                                                                                                      |
| 2025 | <b>Cao, D., Crema, E. R., Lü, H., Yuan, H., &amp; Pomeroy, E. Bayesian models reveal spatiotemporal patterns of body size variation in Holocene Chinese populations. Annual Meeting of the Human Osteoarchaeology Committee, Chinese Archaeological Society, Yili, Xinjiang, China. (Podium)</b>                                                                                                                                 |
| 2025 | <b>Cao, D., Crema, E. R., &amp; Pomeroy, E. Tracking long-term and geographic variation in body size among Chinese populations: Combined insights from</b>                                                                                                                                                                                                                                                                       |

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| 2024 | contemporary and archaeological evidence. 94th Annual Meeting of the American Association of Biological Anthropologists, Baltimore, USA. (Podium)<br>Cao, D., Pomeroy, E., Lu, H., Yuan, H., Yang, F., & Li, S. An analysis of commingled human remains from two Bronze to Iron Age sites in West Tibet. 30th Annual Meeting of the European Association of Archaeologists, Rome, Italy. (Podium) |
| 2024 | Cao, D., Liu, X., & Yuan, H. Dental caries as indicators of agricultural practices in the foothills of Neolithic China. 24th European Meeting of the Paleopathology Association, Leiden, Netherlands. (Poster)                                                                                                                                                                                    |
| 2023 | Cao, D., Crema, E. R., & Pomeroy, E. Estimating intra-limb proportions for commingled remains. 23rd Annual Conference of the British Association for Biological Anthropology and Osteoarchaeology, London, UK. (Poster)                                                                                                                                                                           |
| 2023 | Cao, D., Lu, H., Yang, F., Li, S., Wangdui, X., Yuan, H., & Pomeroy, E. Bioarchaeological insights into health and cultural practices on the ancient Tibetan Plateau in Central Asia. 29th Annual Meeting of the European Association of Archaeologists, Belfast, UK. (Poster)                                                                                                                    |
| 2019 | Cao, D., & Yuan, H. Life-course study of Zhao Xin, an official of the Northern Qi Dynasty. Third Annual Meeting of the Human Osteoarchaeology Committee, Chinese Archaeological Society, Qingdao, China. (Poster)                                                                                                                                                                                 |
| 2018 | Cao, D. Dental caries in the Late Neolithic Gaoshan population in southwest China. Osteoarchaeology Session, Second Chinese Archaeological Congress, Chengdu, China. (Podium)                                                                                                                                                                                                                     |

## FIELDWORK EXPERIENCE

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### Excavations

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| 2021 | <b>Sdingchung Cave Site, Zhongba, Xizang (Tibet), China</b><br>Participated in excavation and was responsible for analysis of human skeletal remains. |
| 2016 | <b>Haimenkou Site, Jianchuan, Dali, Yunnan, China</b><br>Participated in excavation and artefact sorting.                                             |

### Surveys and skeletal documentation

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| 2025 | <b>Shuanghe Cave, Suiyang, Zunyi, Guizhou, China</b><br>Field survey.                                                                 |
| 2020 | <b>Shidaqiu Site, Maerkang, Aba, Sichuan, China</b><br>Field survey and in situ sorting of human skeletal remains.                    |
| 2019 | <b>Xincun Site, Wuding, Chuxiong, Yunnan, China</b><br>Documentation and sorting of human skeletal remains.                           |
| 2019 | <b>Ladian Site, Yuanmou, Chuxiong, Yunnan, China</b><br>Field survey and in situ sorting of human skeletal remains.                   |
| 2019 | <b>Jiangxifen Site, Wuding, Chuxiong, Yunnan, China</b><br>Documentation and sorting of human skeletal remains.                       |
| 2018 | <b>Zhijiaozhongxin Site, Fenyang, Shanxi, China</b><br>Field survey, documentation, and collection of human skeletal remains in situ. |
| 2018 | <b>Qingtai Site, Zhengzhou, Henan, China</b><br>Field survey, documentation, and sorting of human skeletal remains.                   |

## TEACHING AND SUPERVISION

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| 2024-2025 | <b>Supervisor, From Data to Interpretation, Department of Archaeology, University of Cambridge</b><br>Coordinator: Dr Enrico R. Crema. Supervised students in data analysis tools, including R, with emphasis on archaeological and anthropological datasets.                               |
| 2023-2024 | <b>Supervisor, Comparative Human Biology, Department of Archaeology, University of Cambridge</b><br>Coordinator: Dr Mark Dyble. Supervised topics in comparative human biology, evolutionary adaptation, life history, and socio-cultural factors influencing human variability and health. |
| 2023-2025 | <b>Demonstrator, Decoding the Skeleton, Department of Archaeology, University of Cambridge</b>                                                                                                                                                                                              |

Coordinator: Dr Emma Pomeroy. Demonstrated skeletal analysis techniques, including osteological sorting, age and sex estimation, anthropometrics, and palaeopathological observation.

2024-2025

**Demonstrator, Archaeology in Action, Department of Archaeology, University of Cambridge**

Coordinator: Dr Simon Stoddart. Demonstrated techniques in sorting and analysing human skeletal remains.

2018-2021

**Student Teaching Assistant, Sichuan University Bioarchaeology Laboratory, Center for Archaeological Science**

Assisted with laboratory management, equipment maintenance, sample sorting, and teaching.

## PUBLIC ENGAGEMENT AND ADDITIONAL EXPERIENCE

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2021

**Speaker, TEDxKids@Chengdu**

“What I Found from Buried Ancient Teeth / 一颗蛀牙来认识古蜀先民”.

2019

**Sichuan University-Harvard University Study Camp on Ancient Shu Civilisation**

Funded by the Fairbank Center for Chinese Studies at Harvard University and the Research Center for Western China at Sichuan University.

## SKILLS AND SOFTWARE

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- Osteological analysis: skeletal sorting, demographic reconstruction, commingled remains, age/sex estimation, anthropometrics, and palaeopathological observation.
- Statistical and computational analysis: R, Bayesian modelling, data visualisation, and reproducible research workflows.
- Spatial analysis: qGIS, basic map making.
- 3D imaging and morphometrics: Artec Studio, 3D Slicer, SlicerMorph, MeshLab, ImageJ.
- Languages: Chinese (native); English (fluent).